



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
II Sem B.Tech. (CBCS) Mid Semester Examinations - I, February 2023

PH1001-1 – ENGINEERING PHYSICS

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each** Unit.*

List of constants: Velocity of light, $c=3 \times 10^8 \text{ ms}^{-1}$, Planck's constant, $h=6.63 \times 10^{-34} \text{ Js}$,
 Electron mass, $m=9.11 \times 10^{-31} \text{ kg}$, Electron charge, $e=1.6 \times 10^{-19} \text{ C}$,
 Boltzmann constant, $k=1.38 \times 10^{-23} \text{ J/K}$, Neutron mass $=1.68 \times 10^{-27} \text{ kg}$.
 Avogadro number, $N_A = 6.023 \times 10^{26} / \text{kg mole}$, $1 \text{ amu} = 1.6738 \times 10^{-27} \text{ kg}$

Unit – I		Marks	BT*	CO*	PO*
1.	a) List the properties of a well behaved wavefunction.	3	L*1	1	1,2
	b) Derive the time independent Schrodinger wave equation for a particle with mass m and energy E .	4	L2	1	1,2
	c) Compare the de Broglie wavelength of an Iron ball of mass 1 kg moving at a speed of 20 m/s and that of an iron atom which detaches from the iron ball and continues to move at the same speed. (Given: mass of the iron atom is 55.845 amu)	3	L3	1	1,2
2.	a) Explain the properties of matter waves.	3	L1	1	1,2
	b) Obtain the energy eigen values and eigen functions of a particle in a box of infinite depth and width L .	4	L2	1	1,2
	c) Compute the probability of finding a particle between 0 and $0.25L$ in a box of width L , for the ground state.	3	L3	1	1,2
Unit – II					
3.	a) Explain any two crystal systems which have unequal sides.	2	L1	2	1,2
	b) Derive the relation between atomic radius and lattice constant and determine the atomic packing factor of a face centered cubic (FCC) lattice by calculating the number of atoms/unit cell.	4	L2	2	1,2
	c) Draw the following planes in a cubic unit cell: (i) (200) , (ii) $(\bar{1}21)$	4	L3	2	1,2
4.	a) Explain the production of X-rays with a neat diagram.	3	L1	2	1,2
	b) Derive an expression for the inter planar spacing in terms of lattice parameter and Miller indices for the case of a cubic crystal system.	4	L2	2	1,2
	c) Calculate the interplanar distance for the following planes, for a FCC crystal whose atomic radius is $1.246 \text{ \AA}$. (i) (200) (ii) (220)	3	L3	2	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
